EX.No.2.1 CREATE A 3D SOLID MODEL FROM ISOMETRIC DRAWING

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AIM

To create a 3D model from the given isometric drawing using the Autodesk Fusion 360 tool.

REQUIREMENTS

1. Autodesk Fusion 360
2. Windows 10
3. 16 GB of RAM (integrated graphics recommended 1 GB or more)
4. 2.5 Mbps or faster download; 500 Kbps or faster upload

PROCEDURE

STEP-1: Start the sketch command

Select Sketch > Create Sketch.

STEP-2: Select the Sketch Plane

Select the Front (XZ) Plane.

STEP-3: Create a Rectangle

1. Select Sketch > Rectangle > Center Rectangle.
2. Hover over the origin (center) of the sketch; the cursor should snap to this location.
3. Click once to begin placing the rectangle.
4. Press the Tab key and enter 100 mm and 60 mm for the width and height respectively.
5. Click again to place the rectangle.
6. Click Finish Sketch after completing.

STEP-4: Extrude Rectangle in XZ Plane

1. Select Extrude.
2. Select the rectangle sketch.
3. Click again to place the value.
4. Type in the value of 30 mm.
5. Press ENTER to accept the value.

STEP-5: Start the sketch command

Select Sketch > Create Sketch.

STEP-6: Select the Sketch Plane

Select the FRONT SURFACE of the rectangular block.

STEP-7: Create a Circle

1. Select Sketch > Circle > Center Circle.
2. Hover over the origin (center) of the sketch; the cursor should snap to this location.
3. Click once to begin placing the circle.
4. Press the Tab key and enter the diameter value Ø50 mm.
5. Click again to place the circle.

6. Click Finish Sketch after completing.

STEP-8: Extrude Circle

1. Select Extrude.
2. Select the circle sketch.
3. Click again to place the value.
4. Type in the value of 20 mm.
5. Press ENTER to accept the value.

STEP-9: Start the sketch command

Select Sketch > Create Sketch.

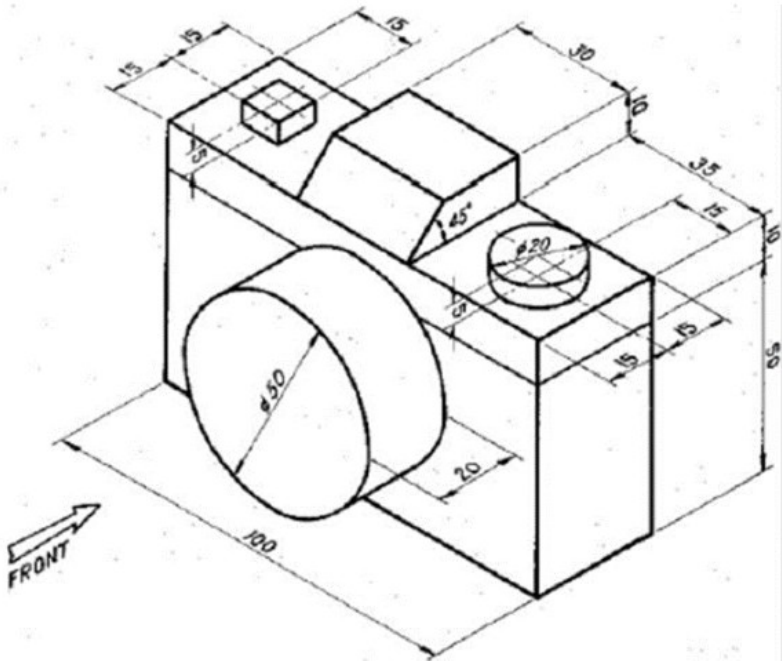
STEP-10: Select the Sketch Plane

Select the TOP SURFACE of the rectangle.

STEP-11: Create a Line

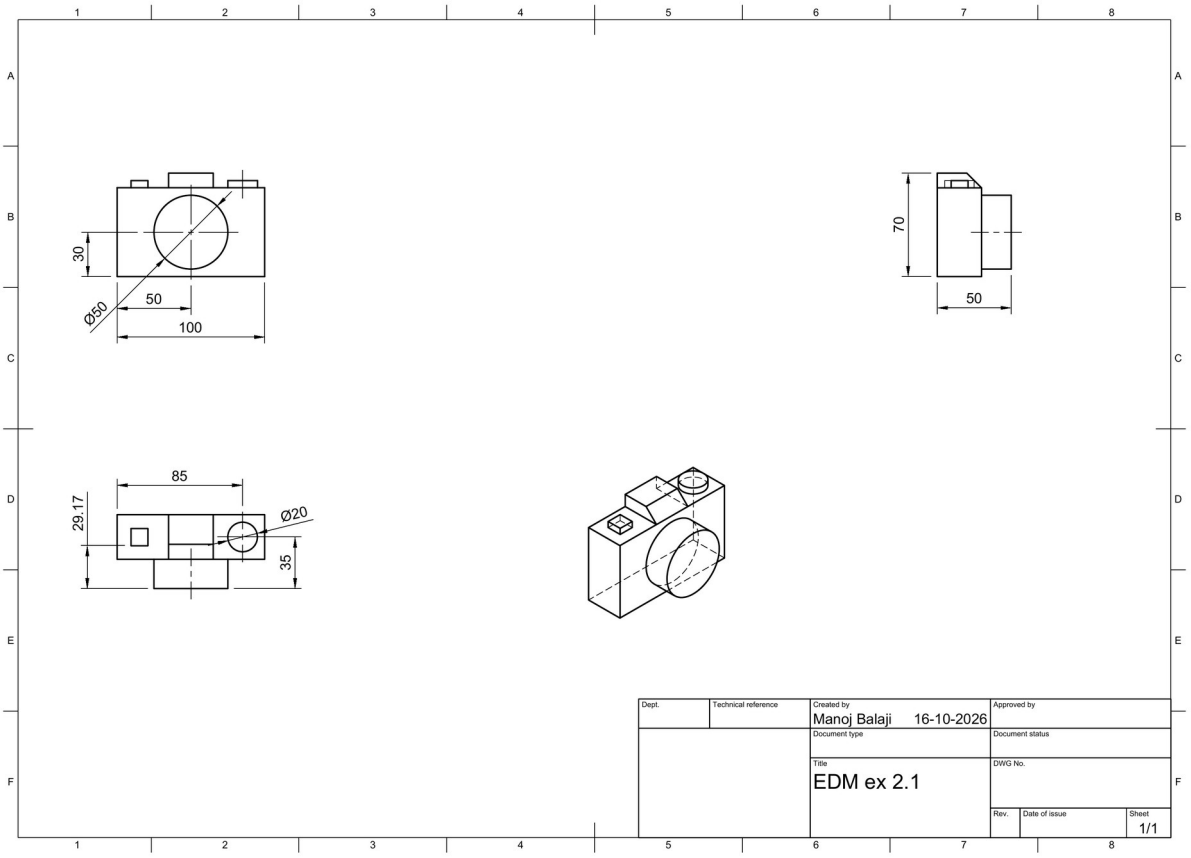
1. Select the Line command.
2. Draw the projections as shown in the diagram (the 45° sloped block, the small square block and the Ø20 circular knob, using the dimensions given).
3. Click Finish Sketch.
4. Extrude the sketch.

INPUT



Given isometric drawing

OUTPUT



3D model drawing sheet: front, top, side and isometric views

RESULT

Thus, the 3D model is created from the given isometric drawing using Autodesk Fusion 360.